

Electronic Instrumentation

Course Code: 2081

Semester: 2

Credits: 3

Dept: Instrumentation Engineering

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Course Overview

Objectives

- Introduce performance characteristics of instruments.
- Learn concepts of PMMC, Moving Iron, and Multimeters.
- Familiarize with DC and AC bridges.
- Understand calibration and working of Wattmeters/Energy meters.
- Study display devices (CRO, DSO, LED, LCD) and recorders.

Teaching Scheme

Lecture	3 Hours/Week
Tutorial	0 Hours/Week
Practical	0 Hours/Week
Total Periods	45 Periods/Semester

Course Outcomes (CO)

CO No.	Description	Cognitive Level
CO1	Identify the performance characteristics of instruments.	Applying
CO2	Utilize moving coil meter as multimeter.	Applying
CO3	Explain the construction and working of watt meters, energy meters and AC & DC Bridges.	Understanding
CO4	Explain the working of display devices – CRO, DSO, LED & LCD, Strip chart recorders and XY recorders.	Understanding

Module 1: Introduction to Instrumentation Engineering

12 Hours

1.1 Generalized Instrumentation System

An instrumentation system is a collection of instruments and auxiliary equipment used to measure, monitor, or control a physical variable. The generalized block diagram consists of:

- 1. **Primary Sensing Element:** Interacts with the measurand (e.g., Thermocouple).
- 2. **Variable Conversion Element:** Converts the output of the sensor into a more suitable form (e.g., Bridge circuit).
- 3. **Variable Manipulation Element:** Changes the level of the signal (e.g., Amplifier).
- 4. **Data Transmission Element:** Transmits data from one stage to another.
- 5. **Data Presentation Element:** Displays or records the value (e.g., Digital display, Recorder).

Malayalam support:

ഒരു അളവുകോലിന്റെ പ്രവർത്തനം മനസ്സിലാക്കുന്നതിനായി, അളവുകോലിന്റെ പ്രവർത്തനം (അളവുകോലിന്റെ പ്രവർത്തനം), അളവുകോലിന്റെ പ്രവർത്തനം. അളവുകോലിന്റെ പ്രവർത്തനം (Measurand) അളവുകോലിന്റെ പ്രവർത്തനം അളവുകോലിന്റെ പ്രവർത്തനം അളവുകോലിന്റെ പ്രവർത്തനം.

1.2 Performance Characteristics

Static Characteristics

These refer to characteristics of instruments measuring quantities that do not vary with time.

- **Accuracy:** Closeness to the true value.
- **Precision:** Consistency or repeatability of measurements.
- **Linearity:** The ability of the instrument to provide an output that is directly proportional to the input.
- **Sensitivity:** Ratio of change in output to change in input.
- **Resolution:** The smallest change in input that can be detected.
- **Hysteresis:** Difference in readings when the input is increased vs decreased.
- **Drift:** Gradual shift in the instrument output over time.

Dynamic Characteristics

These describe the instrument's behavior when measuring rapidly changing quantities.

- **Speed of Response:** How fast the system responds to changes.
- **Fidelity:** The degree to which the system reproduces the input signal without distortion.
- **Lag:** Delay in the response of the instrument.
- **Dynamic Error:** Difference between true value and measured value under dynamic conditions.

1.3 Types of Errors

1. **Gross Errors:** Human mistakes (wrong reading, improper recording).
2. **Systematic Errors:**
 - Instrumental: Due to friction, aging, or faulty construction.
 - Environmental: Due to external conditions like temperature, pressure, humidity.
 - Observational: Parallax error.
3. **Random Errors:** Unpredictable errors that remain after systematic errors are removed.

Module 2: Moving Coil, Moving Iron & Multimeters

9 Hours

2.1 PMMC Instruments

PMMC (Permanent Magnet Moving Coil): Works on the principle that a current-carrying conductor placed in a magnetic field experiences a force (Motor principle).

Torque Equation: $T_d = N \cdot B \cdot I \cdot A$

- N = Number of turns
- B = Flux density
- I = Current
- A = Area of the coil

PMMC instruments are used for **DC measurements only** because the torque reverses if the current reverses.

Malayalam support:

PMMC ഉപകരണങ്ങൾ DC അളവുകൾ അളക്കാനായി ഉപയോഗിക്കുന്നു. ഈ ഉപകരണങ്ങൾ പ്രവർത്തിക്കുന്നത് വൈദ്യുത ചാലകം ഒരു കാന്തീയ മണ്ഡലത്തിൽ സ്ഥിതി ചെയ്യുമ്പോൾ അത് അനുഭവിക്കുന്ന ബലം (Force) അനുസരിച്ചാണ്.

2.2 PMMC as Ammeter and Voltmeter

Ammeter (Shunt)

To measure higher currents, a low resistance (Shunt) is connected in parallel with the meter.

$$R_{sh} = R_m / (m - 1)$$

Where $m = I / I_m$ (Multiplying factor).

Voltmeter (Multiplier)

To measure higher voltages, a high resistance (Multiplier) is connected in series with the meter.

$$R_s = R_m (m - 1)$$

Where $m = V / V_m$.

2.3 Digital Multimeter (DMM)

A DMM converts analog signals to digital form for display. Key blocks include:

- **Input Scaling:** Attenuators for voltage, shunts for current.
- **Rectifier:** Converts AC to DC.
- **Analog to Digital Converter (ADC):** The heart of the DMM.
- **Digital Display:** Usually LCD or LED.

Module 3: Bridges, Wattmeter & Energy Meter

11 Hours

3.1 DC Bridges (Wheatstone Bridge)

Used for measuring medium resistance. At balance, no current flows through the galvanometer.

Balance Condition: $P/Q = R/S$ or $R = (P/Q) \cdot S$

3.2 AC Bridges

Used to measure Inductance (L) and Capacitance (C). Balance occurs when magnitude and phase are both matched.

- **Maxwell Bridge:** Measures medium Q-factor inductance.
- **Hay Bridge:** Measures high Q-factor inductance.
- **Schering Bridge:** Measures unknown capacitance and dissipation factor.
- **Wien Bridge:** Measures frequency and capacitance.

3.3 Wattmeter and Energy Meter

Dynamometer Wattmeter: Consists of fixed coils (Current Coils) and a moving coil (Pressure Coil). Deflection is proportional to $V \cdot I \cdot \cos(\phi)$.

Induction Energy Meter: Measures total energy consumed over time. Uses a rotating aluminum disc. The speed of the disc is proportional to power, and total revolutions are proportional to energy.

Module 4: CRO, DSO, Display Devices & Recorders

11 Hours

4.1 Cathode Ray Oscilloscope (CRO)

Used to visualize waveforms. Components:

- **CRT (Cathode Ray Tube):** Generates the electron beam.
- **Vertical Amplifier:** Amplifies the input signal.
- **Horizontal Sweep Generator:** Creates the time base (Sawtooth wave).
- **Trigger Circuit:** Synchronizes the sweep with the input signal.

4.2 Digital Storage Oscilloscope (DSO)

Unlike CRO, DSO digitizes the input signal using an ADC and stores it in memory. This allows for pre-trigger viewing and permanent storage of waveforms.

4.3 Display Devices & Recorders

- **LED:** Light Emitting Diode. Used in 7-segment and dot-matrix displays.
- **LCD:** Liquid Crystal Display. Consumes very low power.
- **Strip Chart Recorder:** Records a variable against time on a moving paper strip. Uses a potentiometric system for high accuracy.
- **XY Recorder:** Records one variable against another (e.g., V-I characteristics).

Formula Bank & Calculations

Example 1: Ammeter Shunt

Given: Meter resistance $R_m = 100 \text{ } \Omega$, Full-scale current $I_m = 1 \text{ mA}$. Goal: Measure 100 mA .

Calculation:

$$m = I / I_m = 100 / 1 = 100$$

$$R_{sh} = R_m / (m - 1) = 100 / (100 - 1) = 100 / 99 \approx 1.01 \Omega$$

Answer: A shunt resistance of 1.01Ω is required.

Example 2: Voltmeter Multiplier

Given: Meter resistance $R_m = 50 \Omega$, Full-scale current $I_m = 2 \text{ mA}$. Goal: Measure 10 V .

Calculation:

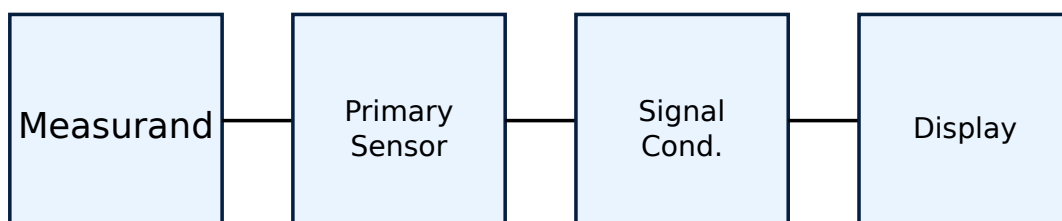
$$V_m = I_m \cdot R_m = 0.002 \cdot 50 = 0.1 \text{ V}$$

$$m = V / V_m = 10 / 0.1 = 100$$

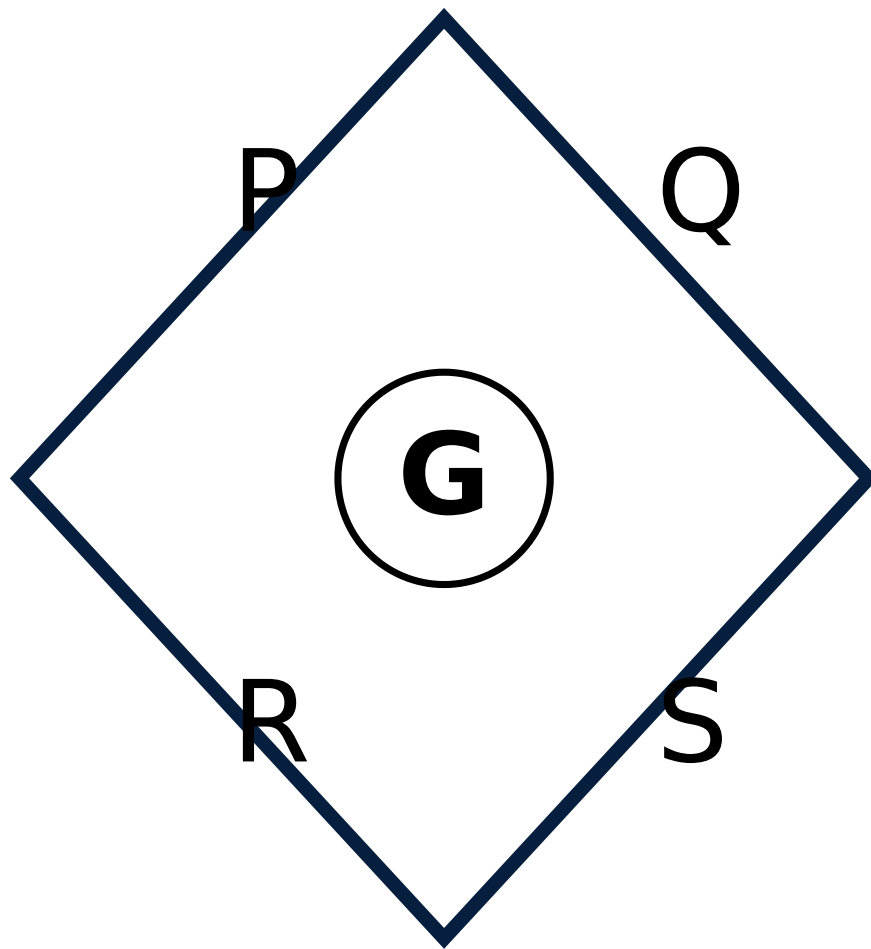
$$R_s = R_m(m - 1) = 50(100 - 1) = 50 \cdot 99 = 4950 \Omega$$

Answer: A series multiplier of 4950Ω is required.

Visual Library



Generalized Block Diagram of an Instrumentation System



Wheatstone Bridge Circuit Diagram

Module-wise Practice Questions

Module 1

Define Accuracy and Precision. (3 Marks)

Accuracy: The closeness of a measured value to the true value of the quantity.

Precision: The degree of repeatability or consistency of a set of measurements. A precise instrument gives the same reading for the same input every time.

Module 2

Why is PMMC not used for AC measurements? (3 Marks)

In AC, the current reverses direction every half cycle. Since the torque in PMMC is proportional to current ($T \propto I$), the torque also reverses direction. Due to the inertia of the moving system, the pointer cannot follow these rapid reversals and stays at zero.

Module 3

Compare Maxwell and Hay Bridges. (5 Marks)

Maxwell Bridge is used for measuring medium Q-factor coils ($1 < Q < 10$). Hay Bridge is modified to measure high Q-factor coils ($Q > 10$). In Hay bridge, the resistance and capacitance are in series, whereas in Maxwell, they are in parallel.

Module 4

What are the advantages of DSO over CRO? (5 Marks)

1. Waveforms can be stored in memory indefinitely.
2. Allows pre-trigger viewing (seeing what happened before the trigger).
3. Waveforms can be transferred to a PC for analysis.
4. Can capture non-repetitive (transient) signals easily.

Practice Model Paper

(Based on REV2021 Pattern)

Part A (Short Answer - 3 Marks each)

1. Define Sensitivity and Resolution.
2. List three types of systematic errors.
3. What is the function of a time-base generator in CRO?
4. State the balance condition for a Wheatstone bridge.

Part B (Medium Answer - 5 Marks each)

1. Explain the construction of a PMMC instrument with a diagram.
2. How can a PMMC meter be converted into a multi-range voltmeter?

3. Explain the working of a Strip Chart Recorder.

Part C (Long Answer - 10 Marks each)

1. Draw the block diagram of a CRO and explain the function of each block.
2. Explain the construction and working principle of an induction type single-phase energy meter.

Quick Revision

- **Static:** Accuracy, Precision, Sensitivity, Linearity, Hysteresis.
- **Dynamic:** Fidelity, Speed of Response, Lag.
- **Errors:** Gross (Human), Systematic (Instrument/Env), Random.
- **PMMC:** DC only, Linear scale, $T_d = \text{NBIA}$.
- **Moving Iron:** AC & DC, Non-linear scale.
- **Bridges:** Wheatstone (R), Maxwell/Hay (L), Schering (C), Wien (f).
- **CRO:** Visualizes waveforms using CRT and time-base.
- **DSO:** Digital storage, memory, ADC.

Glossary

Measurand

The physical quantity or variable being measured.

Calibration

The process of comparing an instrument's reading with a standard to check accuracy.

Parallax Error

An observational error caused by looking at the pointer from an angle.

Shunt

A low resistance connected in parallel to increase ammeter range.

Multiplier

A high resistance connected in series to increase voltmeter range.

Official Source Declaration

This content is architected based on the **SITTTR Kerala REV2021 Syllabus** for Course Code **2081 (Electronic Instrumentation)**.

ഇലക്ട്രോണിക്സ് പഠന സാധനങ്ങൾ / Malayalam learning support

ഇലക്ട്രോണിക്സ് പഠന സാധനങ്ങൾ പഠന സാധനങ്ങൾ പഠന സാധനങ്ങൾ (Electronic Instruments). ഇവയ്ക്ക് പഠന സാധനങ്ങൾ പഠന സാധനങ്ങൾ:

- പഠന സാധനങ്ങൾ പഠന സാധനങ്ങൾ (Accuracy, Precision).
- PMMC, Moving Iron പഠന സാധനങ്ങൾ - ഇവ പഠന സാധനങ്ങൾ പഠന സാധനങ്ങൾ പഠന സാധനങ്ങൾ.
- പഠന സാധനങ്ങൾ (Bridges) - പഠന സാധനങ്ങൾ, പഠന സാധനങ്ങൾ, പഠന സാധനങ്ങൾ പഠന സാധനങ്ങൾ പഠന സാധനങ്ങൾ.
- CRO & DSO - പഠന സാധനങ്ങൾ പഠന സാധനങ്ങൾ പഠന സാധനങ്ങൾ പഠന സാധനങ്ങൾ.